Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **2B** **2026** Fall/Spring/Summer I /**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	2	4								6	
1b	2	2	2							6	
1c	5									5	
2a	6									6	
2b	4	2								6	
2c	5									5	
3a	1	1	1	1	1	1				6	
3b	6									6	
3c	5									5	
								Total		51	

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \|, \perp, \backslash, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by straight lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

Instructions:

R **Do not use decimals anywhere.**

R Do not make any extraneous marks anywhere on the exam.

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in any set in the above. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work.

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points for otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

Use these instructions for the remainder of the exam.

Your J-number $JN:=$ 00499317 $:= (J_k | k \in 1..8)$

Hence

$J_1 = 0$ $J_2 = 0$ $J_3 = 4$ $J_4 = 9$

$J_5 = 9$ $J_6 = 3$ $J_7 = 1$ $J_8 = 7$

Generate your connective list using the following Table, as illustrated in the example below:

Table

0	1	2	3	4	5	6	7	8	9
∨	∧	⟶	↔	∨	∧	⟶	↔	∨	∧

J-number	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8
My J-values	0	0	4	9	9	3	1	7
My C-values	∨	∨	∨	∧	∧	↔	∧	↔
My Connective list	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8

My Connective List

$C_1 = \vee$ $C_2 = \vee$ $C_3 = \vee$ $C_4 = \wedge$

$C_5 = \wedge$ $C_6 = \leftrightarrow$ $C_7 = \wedge$ $C_8 = \leftrightarrow$

Example

The following table is worked out for the J-Number: J00230711

It illustrates how one obtains a list of connectives from one's J number.

The algorithm appears on the following page.

Algorithm:

(0) In the second row, fill in the J-values.

(1) In the third row, **use the entries in the Table in the above** to replace each J-value by its corresponding connective, that is, C-value.

(2) The last row is the list of connectives that you will use for this exam.

J-number	J_1	J_2	J_3	J_4	J_5	J_6	J_7	J_8
J-values	0	0	2	3	0	7	1	1
C-values	∨	∨	⟶	⟷	∨	⟷	∧	∧
Connective list	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8

Example

Algorithm: Finding the value of a formula using one's C-list.

0 To find the value of a formula written in terms of members of a C-list, replace each C-value by the corresponding connective as shown in this example **using your own C-list**.

The formula in the below has been evaluated using the C-list in the example in the above:

$$\left(p\ C_3(\neg q)\right) C_5 \left((\neg p) C_5 q\right)$$

$$= \left(p \longrightarrow (\neg q)\right) \vee \left((\neg p) \vee q\right)$$

R **All formulas on this test must be generated by this method**.

R Use the next algorithm to generate your Truth-Value Number

Truth-Value Number

Your J-number = 00499317

Keep every 0 in your J-number and replace every non-zero number in your J-number by 1 to get

Your (truth-value) **T-number** := 00111111

Define $\forall k \in 1..8$,

T_k := the k[th] entry in your **truth-value J-number** counting from the left. Therefore,

$T_1 := 0$

$T_2 := 0$

$T_3 := 1$

$T_4 := 1$

$T_5 := 1$

$T_6 := 1$

$T_7 := 1$

$T_8 := 1$

6 (1a) Find the truth-tables for after **filling in the blanks** appropriately:

2 (i) $\varphi_{1a(i)} \coloneqq \left(p\ C_1(\neg q)\right)\ C_2\ \left((\neg p)C_3 q\right) = \left(p\ \vee\ (\neg q)\right) \vee\ \left((\neg p)\ \vee\ q\right)$

p	q	$\neg p$	$\neg q$	$p\ C_1(\neg q)$	$(\neg p)C_3 q$	$\left(p\ C_1(\neg q)\right)\ C_2\left((\neg p)C_3 q\right)$
p	q	$\neg p$	$\neg q$	$p\ \vee\ (\neg q)$	$(\neg p)\ \vee\ q$	$\left(p\ \vee\ (\neg q)\right)\ \vee\ \left((\neg p)\ \vee\ q\right)$
0	0	1	1	1	1	1
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	1	0	0	1	1	1

4 (ii) $\left(pC_4 q\right)C_5\ \left(qC_6 r\right) = \left(p \wedge q\right) \wedge\ \left(q \leftrightarrow r\right)$

p	q	r	$pC_4 q$	$qC_6 r$	$(pC_4 q)C_5(qC_6 r)$
p	q	r	$p\ \wedge\ q$	$q \leftrightarrow r$	$(p \wedge q)\ \wedge\ (q \leftrightarrow r)$
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	0	1	0
1	0	0	0	1	0
1	0	1	0	0	0
1	1	0	1	0	0
1	1	1	1	1	1

6 (1b) Find truth-tables for the following after **filling in the blanks** appropriately:

2 (i) $LS \coloneqq \left(\left(pC_7q \right) C_8r \right) = \left(\left(p \ \wedge \ q \right) \leftrightarrow \ r \right)$,

2 (ii) $RS \coloneqq \left(\left(pC_8r \right) \ C_7 \ \left(qC_8r \right) \right) = \left(\left(p \ \leftrightarrow \ r \right) \ \wedge \ \left(q \ \leftrightarrow \ r \right) \right)$, and

2 (iii) decide, if $LS \coloneqq \left(\left(pC_7q \right) C_8r \right) \Leftarrow\Rightarrow \left(\left(pC_8r \right) \ C_7 \ \left(qC_8r \right) \right) =: RS$

p	q	r	$p \ C_7 q$	$(pC_7q)C_8r$	pC_8r	qC_8r	$(pC_8r)C_7(qC_8r)$	$LS \Leftarrow\Rightarrow \ RS$
p	q	r	$p \wedge q$	$(p \wedge q) \leftrightarrow r$	$p \leftrightarrow r$	$q \leftrightarrow r$	$(p \leftrightarrow r) \ \wedge \ (q \leftrightarrow r)$	$LS \Leftarrow\Rightarrow \ RS$
0	0	0	0	1	1	1	1	1
0	0	1	0	0	0	0	0	1
0	1	0	0	1	1	0	0	0
0	1	1	0	0	0	1	0	1
1	0	0	0	1	0	1	0	0
1	0	1	0	0	1	0	0	1
1	1	0	1	0	0	0	0	1
1	1	1	1	1	1	1	1	1

$LS \coloneqq \left(\left(p \wedge q \right) \leftrightarrow r \right) \Leftarrow\Rightarrow \left(\left(p \leftrightarrow r \right) \ \wedge \ \left(q \leftrightarrow r \right) \right) =: RS \quad Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$

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5 (1c) Decide, after **filling in the blanks** appropriately, if:

$$Vld\left(\frac{\left(\quad\right)C_8 r}{\left(pC_8 r\right)C_7\left(qC_8 r\right)}\right) \qquad Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$$

My problem:

$$Vld\left(\frac{\left(p \ \wedge \ q\right) \hookleftarrow r}{\left(p \leftrightarrow r\right) \ \wedge \left(q \leftrightarrow r\right)}\right) \qquad Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$$

6 (2a) Find $\varphi(p,q,r)$ in terms of p,q,r, and connectives, if $\varphi(p,q,r)$ is to have the following truth table:

p	q	r	$\varphi(p,q,r)$	φ_1	φ_2	φ_3	φ_4	φ_5	φ_6	φ_7	φ_8
0	0	0	$T_1 = 0$	0	0	0	0	0	0	0	0
0	0	1	$T_2 = 0$	0	0	0	0	0	0	0	0
0	1	0	$T_3 = 1$	0	0	0	0	0	1	0	0
0	1	1	$T_4 = 1$	0	0	0	0	1	0	0	0
1	0	0	$T_5 = 1$	0	0	0	1	0	0	0	0
1	0	1	$T_6 = 1$	0	0	1	0	0	0	0	0
1	1	0	$T_7 = 1$	0	1	0	0	0	0	0	0
1	1	1	$T_8 = 1$	1	0	0	0	0	0	0	0

$$\varphi(p,q,r)$$

$$= \varphi_1(p,q,r) \vee \varphi_2(p,q,r) \vee \varphi_3(p,q,r) \vee \varphi_4(p,q,r) \vee \varphi_5(p,q,r) \vee \varphi_6(p,q,r)$$

$$= (p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge r) \vee (\neg p \wedge q \wedge \neg r)$$

6 (2b) **Show work on pages that follow**, properly numbered, to find, **using the algebra of propositions**, and the **identities expressing $p \longrightarrow q$ and $p \longleftrightarrow q$ in terms of the connectives: ¬, ∨, and ∧** to determine the equivalent expressions for $\psi(p,q)$ as a disjunction of (conjunction of literals) (called DNF) and as a conjunction of (disjunction of literals) (called CNF). Cross out the disjuncts, respectively conjuncts, from the disjunctions and conjunctions provided in the lines marked DNF and CNF respectively so that equality holds.

4 (i)

$$\psi(p,q)$$

$$:= \quad \neg\left(\left(p\ C_1(\neg q)\right)\ C_2\ \left((\neg p) C_3 q\right)\right)$$

$$= \quad \neg\left(\left(p\ \vee\ (\neg q)\right)\ \vee \left((\neg p)\ \vee\ q\right)\right)$$

$$= \quad \cancel{\left(p \wedge q\right)} \vee \cancel{\left(p \wedge (\neg q)\right)} \vee \cancel{\left((\neg p) \wedge q\right)} \vee \cancel{\left((\neg p) \wedge (\neg q)\right)} \text{(DNF)}$$

$$= \quad \cancel{\left(p \vee q\right)} \wedge \cancel{\left(p \vee (\neg q)\right)} \wedge \cancel{\left((\neg p) \vee q\right)} \wedge \cancel{\left((\neg p) \vee (\neg q)\right)} \text{(CNF)}$$

2 (ii) Fill in the column for $\psi(p,q)$, **using ONLY the columns of conjuncts (respectively disjuncts) as needed**, on the basis of your answers to part (2b)(i). Fill in the last column from (1a)(i). **Wherever $\boldsymbol{\psi(p,q)}$ has 0, $\boldsymbol{\varphi_{1a(i)}}$ should have 1 and vice versa.** If this is not the case, something is wrong and **0 points will be awarded for both (1a)(i) and (2b)**.

p	q	$\neg p$	$\neg q$	$p \wedge q$	$p \wedge (\neg q)$	$(\neg p) \wedge q$	$p \wedge (\neg q)$	$(\neg p) \wedge (\neg q)$	$\psi(p,q)$	$\varphi_{1a(i)}$
0	0	1	1	0	0	0	0	1	0	1
0	1	1	0	0	0	1	0	0	0	1
1	0	0	1	0	1	0	1	0	0	1
1	1	0	0	1	0	0	0	0	0	1

p	q	$\neg p$	$\neg q$	$p \vee q$	$p \vee (\neg q)$	$(\neg p) \vee q$	$p \vee (\neg q)$	$(\neg p) \vee (\neg q)$	$\psi(p,q)$	$\varphi_{1a(ii)}$
0	0	1	1	0	1	1	1	1	0	1
0	1	1	0	1	0	1	0	1	0	1
1	0	0	1	1	1	0	1	1	0	1
1	1	0	0	1	1	1	1	0	0	1

2b

$\neg((p \vee (\neg q)) \vee ((\neg p) \vee q))$

$=((\neg(p \vee (\neg q))) \wedge (\neg((\neg p) \vee q)))$ $(\neg(p \vee q) = (\neg p) \wedge (\neg q))$

$=(((\neg p) \wedge (\neg(\neg q))) \wedge ((\neg(\neg p)) \wedge (\neg q)))$ $(\neg(p \vee q) = (\neg p) \wedge (\neg q))$

$=(((\neg p) \wedge q) \wedge (p \wedge (\neg q)))$ $(\neg(\neg p) = p)$

$=((p \wedge (\neg p)) \wedge (q \wedge (\neg q)))$ $((p \wedge q) \wedge r = p \wedge (q \wedge r))$

$=((\bot) \wedge (\bot))$ $(p \wedge (\neg p) = \bot)$

$=\bot$ $(p \wedge p = p)$

$=\bot$ (DNF)

$=\bot$ (CNF)

5 (2c) Show work on pages that follow, properly numbered, to decide, **using the algebra of propositions**, if the following inference is valid and circle the appropriate choice:

$$Vld \quad \frac{\left(pC_3q\right)\rightarrow\left(p\right) \quad \left(pC_3q\right)\rightarrow q}{\left(\left(pC_3q\right)\rightarrow\left(pC_5q\right)\right)} \qquad \textcircled{Y} \quad N \quad \textcircled{Pf} \quad W$$

My problem:

$$Vld \quad \frac{\left(p \vee q\right)\rightarrow\left(p\right) \quad \left(p \vee q\right)\rightarrow q}{\left(\left(p \vee q\right)\rightarrow\left(p \wedge q\right)\right)} \qquad \textcircled{Y} \quad N \quad \textcircled{Pf} \quad W$$

2c

$(P \to R) \wedge (P \to S)$

$= ((\neg P) \vee R) \wedge ((\neg P) \vee S)$ $\quad (p \to q = (\neg p) \vee q)$

$= (\neg P) \vee (R \wedge S)$ $\quad ((\neg P \vee R) \wedge (\neg P \vee S) = \neg P \vee (R \wedge S))$

$= P \to (R \wedge S)$ $\quad ((\neg p) \vee q = p \to q)$

$P := (p \vee q), R := p, S := q$

$((p \vee q) \to p) \wedge ((p \vee q) \to q) = (p \vee q) \to (p \wedge q)$

6 (3a) Show work on **pages that follow**, properly numbered, to decide if the following proposition, denoted as indicated is a tautology, contradiction, or contingency, and **circle an appropriate answer**.

1 (i) $\alpha \coloneqq \Big((\top\, C_1 p) C_3\, \bot\Big) = \Big((\top \vee p) \vee \bot\Big)$

$\textcircled{Tlg(\alpha)}$ $Cdn(\alpha)$ $Cng(\alpha)$

1 (ii) $\beta \coloneqq \Big(\top\, C_2 (p C_4\, \bot)\Big) = \Big(\top \vee (p \wedge \bot)\Big)$

$\textcircled{Tlg(\beta)}$ $Cdn(\beta)$ $Cng(\beta)$

1 (iii) $\gamma \coloneqq \Big((\bot\, C_3 p) C_5\, \top\Big) = \Big((\bot \vee p) \wedge \top\Big)$

$Tlg(\gamma)$ $Cdn(\gamma)$ $\textcircled{Cng(\gamma)}$

1 (iv) $\delta \coloneqq \Big(\bot\, C_4 (p C_6\, \top)\Big) = \Big(\bot \wedge (p \leftrightarrow \top)\Big)$

$Tlg(\delta)$ $\textcircled{Cdn(\delta)}$ $Cng(\delta)$

1 (v) $\zeta \coloneqq \Big(\top\, C_5 (p C_7\, \bot)\Big) = \Big(\top \wedge (p \wedge \bot)\Big)$

$Tlg(\zeta)$ $\textcircled{Cdn(\zeta)}$ $Cng(\zeta)$

1 (vi) $\eta \coloneqq \Big((\top\, C_6 p) C_8\, \bot\Big) = \Big((\top \leftrightarrow p) \leftrightarrow \bot\Big)$

$Tlg(\eta)$ $Cdn(\eta)$ $\textcircled{Cng(\eta)}$

3a.i	3a.ii	3a.iii
$\alpha := (\top \vee p) \vee \bot$ $= \top \vee \bot \quad (\top \vee p = \top)$ $= \top$ $Tlg((\top \vee p) \vee \bot)$	$\beta := \top \vee (p \wedge \bot)$ $= \top \vee \bot \quad (p \wedge \bot = \bot)$ $= \top$ $Tlg(\top \vee (p \wedge \bot))$	$\gamma := (\bot \vee p) \wedge \top$ $= p \wedge \top \quad (\bot \vee p = p)$ $= p$ $Cng((\bot \vee p) \wedge \top)$
3a.iv	3a.v	3a.vi
$\delta := \bot \wedge (p \leftrightarrow \top)$ $= \bot \wedge p \quad (p \leftrightarrow \top = p)$ $= \bot$ $Cdn(\bot \wedge (p \leftrightarrow \top))$	$\zeta := \top \wedge (p \wedge \bot)$ $= \top \wedge \bot \quad (p \wedge \bot = \bot)$ $= \bot$ $Cdn(\top \wedge (p \wedge \bot))$	$\eta := (\top \leftrightarrow p) \leftrightarrow \bot$ $= p \leftrightarrow \bot \quad (\top \leftrightarrow p = p)$ $= \neg p$ $Cng((\top \leftrightarrow p) \leftrightarrow \bot)$

6 (3b) Show work on pages that follow, properly numbered, to decide, **using axioms and rules of inference**, if the following inference is valid and circle the appropriate choice. If (2c) has been submitted, then the answer to (3b) must agree with the answer of (2c); otherwise, 0 points on both.

$$Vld \quad \frac{\left(pC_3q\right)\rightarrow\left(p\right) \quad \left(pC_3q\right)\rightarrow q}{\left(\left(pC_3q\right)\rightarrow\left(pC_5q\right)\right)}$$

Ⓨ N Ⓟf W

My problem:

$$Vld \quad \frac{\left(p \vee q\right)\rightarrow\left(p\right) \quad \left(p \vee q\right)\rightarrow q}{\left(\left(p \vee q\right)\rightarrow\left(p \wedge q\right)\right)}$$

Ⓨ N

3b

$$\dfrac{\dfrac{\dfrac{Vld(\frac{r}{r})}{Vld(\frac{(r\to p)\wedge(r\to q)}{(r\to p)\wedge(r\to q)})}\ \langle\, r \longleftarrow (r\to p)\wedge(r\to q)\,\rangle}{Vld(\frac{(r\to p)\wedge(r\to q)}{r\to(p\wedge q)})}\ \langle\, r \longleftarrow (p\vee q)\,\rangle}{Vld(\frac{((p\vee q)\to p)\wedge((p\vee q)\to q)}{(p\vee q)\to(p\wedge q)})}$$

5 (3c) Fill in the gaps and decide if:

$$Vld\left(\frac{p\ C_2\ q \quad qC_4\ r \quad r\ C_6\ s}{p\ C_8\ q}\right) \qquad Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$$

My Problem:

$$Vld\left(\frac{p\ \vee\ q \quad q\ \wedge\ r \quad r \leftrightarrow s}{p\ \leftrightarrow\ q}\right) \qquad Y \quad \textcircled{N} \quad \textcircled{Pf} \quad W$$

3c

$p{:=}0, q{:=}1, r{:=}1, s{:=}1$

$p \vee q = 0 \vee 1 = 1$

$q \wedge r = 1 \wedge 1 = 1$

$r \leftrightarrow s = 1 \leftrightarrow 1 = 1$

$p \leftrightarrow q = 0 \leftrightarrow 1 = 0$